🚚 Smart Transport & Fleet Management System (STFMS)

Language: ANSI C
Architecture: Structured Enterprise Console System
Storage: Binary File-Based Persistence
Version: 1.0

📌 Executive Summary

The Smart Transport & Fleet Management System (STFMS) is a structured transportation enterprise simulation platform developed entirely in ANSI C.

The system integrates core fleet operational components including:

• Secure administrator authentication
• Vehicle fleet management
• Driver management
• Trip recording and monitoring
• Fuel consumption calculation
• Trip cost computation
• Statistical cost analytics
• Timestamp-based audit logging

STFMS models centralized transportation workflow using deterministic structured programming and binary file-based persistence without reliance on external database systems.

🏗️ System Architecture

STFMS follows a layered logical architecture:

• Authentication Layer
• Vehicle Management Module
• Driver Management Module
• Trip Processing Engine
• Trip Cost Analytics Engine
• Audit Logging Layer
• Binary Persistence Layer

Although implemented within a single source file, functional responsibilities are logically separated to simulate enterprise-grade modular design.

🔐 Authentication Module

The system ensures secure administrative access.

Features:

• Automatic admin initialization on first execution
• Password hashing using djb2 algorithm
• No plaintext credential storage
• Binary verification against `transport_users.dat`

Default Credentials:

Username: admin
Password: admin123

Hashing Logic:

```
hash = ((hash << 5) + hash) + character
```

🚚 Vehicle Management Module

Each vehicle record contains:

• Vehicle ID
• Vehicle Model
• Capacity
• Fuel Efficiency (km per liter)
• Maintenance Cost

Functional Capabilities:

• Add vehicle
• Maintain fleet database

Fuel efficiency parameter is used dynamically during trip cost calculation.

Storage File: `vehicles.dat`

👨‍✈️ Driver Management Module

Each driver record contains:

• Driver ID
• Name
• License Number
• Performance Rating (0–5)

Functional Capabilities:

• Add driver
• Maintain driver database

Storage File: `drivers.dat`

🛣️ Trip Processing Engine

The trip engine records operational journeys performed by vehicles and drivers.

Each trip record contains:

• Trip ID
• Vehicle ID
• Driver ID
• Distance (km)
• Fuel Used
• Trip Cost
• Timestamp

Operational Flow:

• Validate vehicle existence
• Capture driver and distance data
• Compute fuel consumption
• Compute trip cost
• Record trip transaction
• Persist trip data

Fuel Consumption Formula:

```
Fuel Used = Distance / Fuel Efficiency
```

Trip Cost Formula:

```
Trip Cost = (Fuel Used × 100) + Maintenance Cost
```

The system integrates both operational fuel cost and fixed maintenance cost into overall trip cost.

Storage File: `trips.dat`

📊 Trip Cost Analytics Engine

The analytics module processes trip records and computes:

• Mean (Average Trip Cost)
• Variance
• Standard Deviation

Formulas Used:

```
Mean = Σ Trip Costs / N
Variance = Σ (Cost − Mean)² / N
Standard Deviation = √Variance
```

Purpose:

• Evaluate operational cost distribution
• Measure fleet cost variability
• Analyze transportation efficiency
• Support cost optimization assessment

📝 Audit Logging

All major system activities are recorded in:

`transport_logs.txt`

Logged Events:

• Admin login
• Vehicle addition
• Driver addition
• Trip recording
• Analytics execution
• System exit

Each log entry includes timestamp metadata for traceability and accountability.

💾 Data Storage Design

Binary files used:

File	Purpose
transport_users.dat	Authentication records
vehicles.dat	Fleet database
drivers.dat	Driver database
trips.dat	Trip transactions
transport_logs.txt	Audit logs

Binary storage ensures:

• Efficient record alignment
• Faster read/write operations
• Reduced storage overhead
• Persistent fleet operational integrity

⚙️ Compilation & Execution

Compile:

```
gcc transport_management_system.c -o transport -lm
```

Run:

```
./transport
```

🧠 Engineering Highlights

• ANSI C structured programming
• Fleet workflow abstraction
• File-based transport database simulation
• Real-time fuel consumption calculation
• Integrated maintenance cost modeling
• Statistical cost analysis
• Deterministic memory management
• Timestamp-based operational traceability

🎯 Project Significance

STFMS demonstrates the capability to design and implement centralized fleet and transportation workflow using low-level systems programming.

It reflects:

• Transportation process modeling
• Structured data persistence design
• Operational cost analytics integration
• Secure authentication handling
• Enterprise-grade fleet management logic

🏁 Conclusion

The Smart Transport & Fleet Management System (STFMS) is a complete transportation enterprise simulation platform implemented in pure ANSI C.

It integrates:

Authentication
Vehicle Management
Driver Management
Trip Processing
Cost Analytics
Audit Logging

All within a structured and deterministic runtime architecture.